

Week 6

October 17, 2025

Exercise 1. Show that $\sum_{j=1}^n (a_j - a_{j-1}) = a_n - a_0$, where $a_0, a_1, \dots, a_n$ is a sequence of real numbers. This type of sum is called **telescoping**.

Exercise 2. Sum both sides of the identity $k^2 - (k-1)^2 = 2k - 1$ from $k = 1$ to $k = n$ and use Exercise 1 to find

- (a) a formula for $\sum_{k=1}^n (2k - 1)$ (the sum of the first n odd natural numbers).
- (b) a formula for $\sum_{k=1}^n k$.

Exercise 3. The following three algorithms sort the input sequence $a_1, \dots, a_n$ of real numbers in ascending order.

Algorithm 1 Bubble Sort	Algorithm 2 Selection Sort	Algorithm 3 Insertion Sort
<pre> for $i = 1$ to $n - 1$ do for $j = 1$ to $n - i$ do if $a_j > a_{j+1}$ then swap a_j and a_{j+1} end if end for end for </pre>	<pre> for $i = 1$ to $n - 1$ do $\text{min} \leftarrow i + 1$ for $j = i + 1$ to n do if $a_{\text{min}} > a_j$ then $\text{min} \leftarrow j$ end if end for if $a_i > a_{\text{min}}$ then swap a_i and a_{min} end if end for </pre>	<pre> for $j = 2$ to n do $i \leftarrow 1$ while $a_j > a_i$ do $i \leftarrow i + 1$ end while $m \leftarrow a_j$ for $k = 0$ to $j - i - 1$ do $a_{j-k} \leftarrow a_{j-k-1}$ end for $a_i \leftarrow m$ end for </pre>

Use Bubble Sort, Selection Sort and Insertion Sort to sort the following sequence:

9, 12, -43, 20, -2, 3, 7, 28, 19.

Exercise 4. Adapt the bubble sort algorithm so that it stops when no swaps are required (when the list is sorted). Express this more efficient version of the algorithm in pseudocode.

Exercise 5. Two strings are anagrams if each can be formed from the other string by rearranging its characters. Devise an algorithm to determine whether two strings are anagrams

Exercise 6. Describe an algorithm that determines whether a function from a finite set to another finite set is injective.

Exercise 7.

$$\begin{array}{ll}
 L_{x_1} = (y_3, y_1, y_2) & L_{y_1} = (x_2, x_1, x_3) \\
 L_{x_2} = (y_2, y_3, y_1) & L_{y_2} = (x_1, x_3, x_2) \\
 L_{x_3} = (y_1, y_2, y_3) & L_{y_3} = (x_3, x_2, x_1)
 \end{array}$$

Let L_x for $x \in X = \{x_1, x_2, x_3\}$ be the preference list of x as given above and let L_y for $y \in Y = \{y_1, y_2, y_3\}$ be the preference list of y as given above.

We say that a matching is X -optimal (resp. Y -optimal) if all elements of X (resp. Y) are matched with their highest preference.

The matching $\{(x_1, y_1), (x_2, y_3), (x_3, y_2)\}$ is

- ☐ unstable.
- ☐ stable and Y -optimal.
- ☐ stable and X -optimal.
- ☐ stable but not a stable matching that is X - or Y -optimal.

Exercise 8. Use the cashier's algorithm to make change using quarters (25 cents), dimes (10 cents), nickels (5 cents), and pennies (1 cent) for:

1. 87 cents.
2. 49 cents.
3. 99 cents.
4. 33 cents.

Exercise 9. Let $\{A, B, C, D\}$ be a set of men, and $\{a, b, c, d\}$ a set of women. We want to match up men and women using the Gale-Shapley algorithm in two different ways. The preferences of men and women are given in the following lists, going from most preferable on the left to least preferable on the right.

Men	1st	2nd	3rd	4th
A	c	d	b	a
B	d	c	a	b
C	a	c	b	d
D	b	d	a	c

Women	1st	2nd	3rd	4th
a	D	A	B	C
b	C	B	A	D
c	C	B	A	D
d	D	A	B	C

1. If the men propose, and women accept/reject, what is the matching after the algorithm terminates?
2. If the women propose, and men accept/reject, what is the matching after the algorithm terminates?
3. Who is the best possible (stable) valid partner for "a"?

Exercise 10. We use Bubble Sort, Selection Sort, and Insertion Sort to sort the following sequence (see Exercise 3):

9, 12, -43, 20, -2, 3, 7, 28, 19.

1. How many comparisons are done in each of the algorithms?
2. How many swaps are done in each of the algorithms?
3. In the worst-case scenario, what is the order of how many comparisons and how many swaps are done by the three algorithms for an input sequence of length $n + 1$?